

# XenaKube\_Study I (2026)

for smart Rubik's cube, virtual cello (SWAM physical model), and live-inscribed score.

COMPOSER / PERFORMER HAN GEUL LEE  
DURATION 7.4 minutes

XenaKube is a solo work in which a stateful smart cube shapes a physically modeled cello-body while the score is inscribed live from the system's performed output. The study follows a scramble → solve arc, which guides the improvisation of the performance.

The cello-body is a SWAM physical model. A turn produces a phrase — pluck cluster, bowed run, sustained line, glissando, tremolo. Phrase length, density, register come from the active K-vertex's parameter triple. Gyro values modulate bow pressure, position, and vibrato continuously.

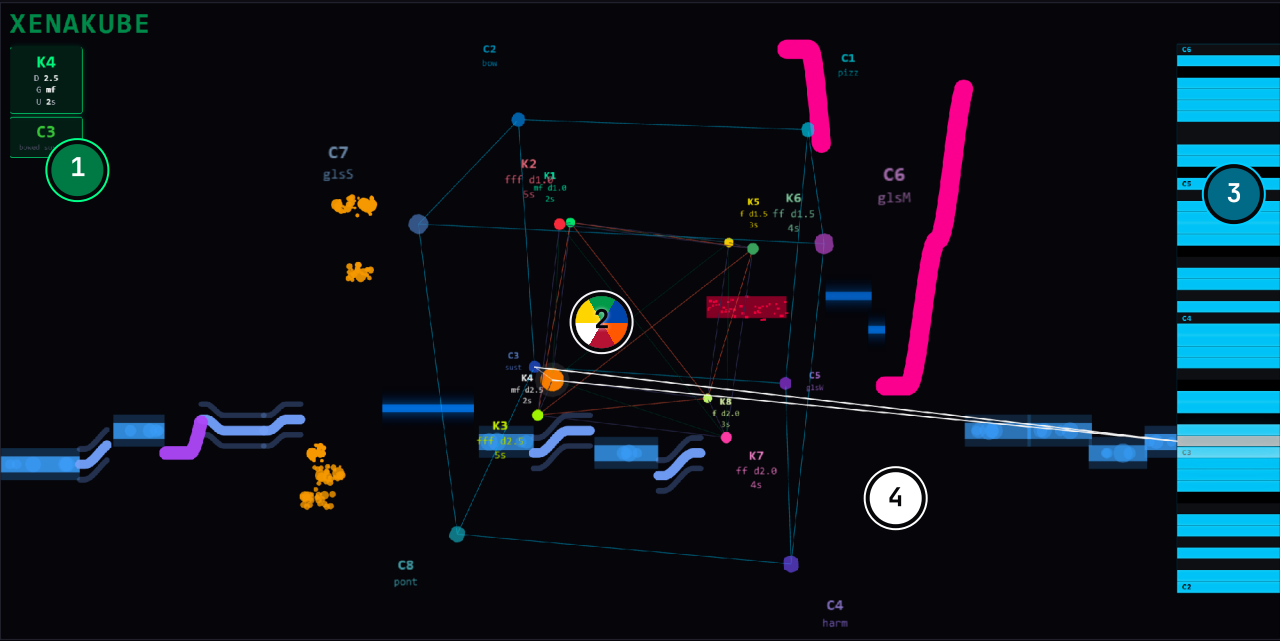
CUBE > RELAY > ENGINE > SWAM > SCORE

GAN i4 smart cube → Web Bluetooth (Chrome) → Node relay → TypeScript engine (S4, K<sub>i</sub>/C<sub>i</sub>, sieve, recognition, expression) → OSC → Max/MSP hosting SWAM Cello 3 → MIDI echo back to browser → live score.

## SOUND-COMPLEXES (C1 - C8)

Eight Xenakian sound-types. Swatch = brush colour as it appears in the rolling score.

- C1** Pizzicato cloud — *plucked grains*
- C2** Bowed asc / desc cloud — *directional sweep*
- C3** Flat bowed cloud — *stationary band*
- C4** Ionised atom — *harmonics + interference*
- C5** Ataxic gliss — *wild, wide intervals*
- C6** Ordered gliss — *stepwise asc / desc*
- C7** Sustained sliding — *tasto, drift-like*
- C8** Sul ponticello tremolo — *near-bridge flicker*



- 1** K · C CARDS  
Active K-vertex (parameter triple) and C-complex (C1–C8).
- 2** LIVE CUBE VIEW  
3D played state with translucent ghost showing the nearest of 24 S4 rotation symmetries. See 'K-vertices'.
- 3** PITCH SIEVE  
49 cells, C2–C6. Lit briefly as pitches sound — an irregular pitch lattice that mutates as the cube moves.
- 4** LINE OVERLAY  
White geometry linking the active cube vertex and ghost vertex to their sieve cell at the right edge where the materials are produced from.

## FACE / TEMPORAL IDENTITY

Each of the 12 face-moves owns a fixed gesture **type**. K<sub>i</sub>/C<sub>i</sub> modulate **content** inside the type; the type is predictable from the face. Primed pairs share an axis, invert articulation.

|    |       |           |    |       |             |
|----|-------|-----------|----|-------|-------------|
| U  | pluck | ↑ attack  | R  | stab  | – attack    |
| U' | fade  | ↓ release | R' | burst | ~ iterative |
| D  | stab  | ↓ attack  | F  | swell | ↑ sustain   |
| D' | drone | – sustain | F' | swell | ↓ sustain   |
| L  | swell | ↑ sustain | B  | pluck | – attack    |
| L' | fade  | ↓ release | B' | drone | ~ sustain   |